

Transfer Learning versus Training from Scratch for Scene Image Classification

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Abstract—This paper presents a controlled, empirical comparison of transfer learning and training from scratch using the ResNet-18 architecture on the six-class Intel Image Classification dataset. To isolate the effects of feature pretraining quality from those of trainable capacity, we define a capacity-matched design featuring three training conditions: (i) *pretrained-frozen*, where ImageNet-pretrained weights are loaded and early layers are frozen; (ii) *random-frozen*, where early layers are randomly initialized and frozen as a capacity-matched control; and (iii) *random-full*, where the entire network is trained from scratch. In all frozen conditions, batch normalization layers in the frozen block are kept in evaluation mode to prevent representation drift. The models are trained with validation-based early stopping across three distinct random seeds. Evaluation on the held-out test split demonstrates that the *pretrained-frozen* model achieves a mean test accuracy of 93.27%, significantly outperforming the *random-frozen* (66.54%) and *random-full* (84.69%) configurations. The results isolate a substantial accuracy benefit (> 50%) directly attributable to the quality of pretrained features under matched capacities, and illustrate the critical role of pretraining in low-compute, small-dataset regimes.

Index Terms—Transfer Learning, Image Classification, Parameter Freezing, Network Capacity, Empirical Analysis.

I. INTRODUCTION

Deep convolutional neural networks (CNNs) have achieved state-of-the-art performance on a wide range of computer vision tasks, but training them from randomly initialized weights usually requires extremely large labeled datasets and substantial compute budgets to avoid overfitting and ensure convergence. To address this, transfer learning has emerged as a standard practice where a model is initialized with weights learned on a massive source dataset (such as ImageNet) and subsequently adapted to a smaller target domain.

A naive comparison between a pre-trained network with frozen early layers and a fully trained network from random initialization confounds two distinct factors: the quality of the pretrained representations, and the reduced trainable parameter count (trainable capacity) of the frozen network.

To resolve this ambiguity, this paper implements a three-condition experimental design to isolate these factors:

- 1) **Pretrained-Frozen (PT-F):** ResNet-18 model initialized with ImageNet weights, where early layers (θ_f) are frozen, and only the final convolutional block (`layer4`) and the classification head (θ_t) are optimized.
- 2) **Random-Frozen (R-F):** A capacity-matched control identical to PT-F, but with θ_f randomly initialized and frozen.

- 3) **Random-Full (R-FL):** The entire ResNet-18 network is initialized randomly and fully trained.

Comparing condition (i) vs (ii) directly isolates the effect of pretraining quality under capacity-matched settings. Comparing (ii) vs (iii) isolates the effect of network capacity under random initialization. We evaluate all conditions on the Intel Image Classification dataset, compiling results across multiple seeds with validation-based early stopping to perform a comprehensive comparison of final accuracy and convergence speed.

II. RELATED WORK

The utility of pre-trained visual representations and the limits of transfer learning have been studied extensively:

He et al. [1] introduced the Deep Residual Learning framework (ResNet), establishing that residual connections allow networks to be trained to much greater depths without optimization degradation. ResNet-18 serves as the standard, lightweight backbone in this work due to its computational feasibility and representative residual structure.

Yosinski et al. [2] investigated the transferability of features in deep neural networks. They proved that first-layer features are general-purpose (e.g., edge and color detectors) regardless of the dataset, while final-layer features are task-specific. Crucially, they observed that freezing early layers maintains general features, but splitting a network mid-way can cause optimization difficulties due to co-adaptation of features in adjacent layers. This motivates our partition boundary, where only the final stage (`layer4`) and head are trainable.

He et al. [3] challenged the necessity of ImageNet pre-training for target tasks in their seminal paper "Rethinking ImageNet Pre-training". They demonstrated that models trained from scratch (random initialization) can match or even exceed the accuracy of pre-trained models on target tasks such as COCO object detection, provided they are trained for long enough. They concluded that pre-training primarily speeds up convergence rather than improving generalization limits when data is sufficient. Our work explores whether these findings hold for scene classification under small data and tight training patience budgets.

III. METHODS

A. Base Architecture and Partitioning

We employ ResNet-18 [1] as our base architecture. Let f_θ denote the network with parameters θ . We partition θ into a

frozen subset θ_f and a trainable subset θ_t . Following findings that early layers learn low-level, generic representations [2], we define θ_f as the input convolution (`conv1`), first batchnorm (`bn1`), and the first three residual stages (`layer1`, `layer2`, `layer3`). The trainable subset θ_t consists of the final residual stage (`layer4`) and the 6-class output layer (`fc`), representing the task-specific adapter.

In total, θ_f contains 2,782,784 frozen parameters, while θ_t contains 8,396,806 trainable parameters (including batchnorm layers in `layer4` and the classification linear weights and bias).

B. Batch Normalization Handling

Batch Normalization (BN) layers keep track of running statistics (mean and variance) during training. If early layers (θ_f) are frozen but their BN layers continue updating running statistics, the internal representation of the network will drift inconsistently, leading to severe optimization decay. To prevent this, we enforce that all BN layers within θ_f are explicitly set to evaluation mode (`model.eval()`) during training. Thus, their running statistics are frozen and they function as deterministic linear transforms matching their initialization.

C. Data Preprocessing and Splitting

The Intel Image Classification dataset [4] contains natural scene images of size 150×150 pixels across 6 classes: *buildings*, *forest*, *glacier*, *mountain*, *sea*, and *street*. Images are preprocessed identically across all conditions:

- 1) Resized such that the shorter side is 224px.
- 2) Center-cropped to 224×224 pixels.
- 3) Normalized per-channel using ImageNet mean $\mu = [0.485, 0.456, 0.406]$ and standard deviation $\sigma = [0.229, 0.224, 0.225]$.

The dataset is partitioned as follows: the standard training split (`seg_train`) containing 14,034 images is divided into a 90% training subset (12,631 images) and a 10% validation subset (1,403 images) carved out using class-stratified sampling to preserve label distributions. The test split (`seg_test`) containing 3,000 images is held out completely and only used for final model evaluation.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

For all three conditions, we optimize the trainable parameters θ_t to minimize the cross-entropy loss:

$$\mathcal{L}(\theta_t) = -\frac{1}{N} \sum_{i=1}^N \log p_{\theta}(y_i | x_i) \quad (1)$$

Training uses SGD with momentum $\beta = 0.9$, weight decay $\lambda = 10^{-4}$, and a learning rate $\eta = 10^{-3}$. We set the batch size to 32.

To ensure fair comparison, validation loss is evaluated at the end of each epoch. We apply early stopping with a patience of 5 epochs: training terminates if validation loss fails to improve for 5 consecutive epochs, and the checkpoint with the lowest

validation loss is saved. All experiments are executed across three seeds: 42, 100, and 2026.

B. Overall Test Performance

Table I summarizes the test accuracy (mean and standard deviation across seeds) and convergence speed (epochs to early stopping) for each condition.

TABLE I
TEST ACCURACY AND CONVERGENCE SPEED COMPARISON

Condition	Mean Test Acc.	Std. Dev.	Mean Epochs
Pretrained-Frozen	93.27%	0.23%	8.0
Random-Frozen	66.54%	4.04%	8.7
Random-Full	84.69%	1.87%	15.0

C. Detailed Class Metrics

For each condition, we identify the seed closest to the mean test accuracy (representative run) to report detailed class-wise metrics. These representative runs correspond to Seed 100 for PT-F, Seed 100 for R-F, and Seed 100 for R-FL.

Table II details the per-class precision, recall, and F1-score for the representative seed of each condition.

TABLE II
PER-CLASS TEST METRICS FOR REPRESENTATIVE RUNS

Class	Pretrained-Frozen			Random-Frozen			Random-Full		
	Prec	Rec	F1	Prec	Rec	F1	Prec	Rec	F1
Buildings	93.2%	93.4%	93.3%	53.0%	64.5%	58.2%	82.5%	82.8%	82.6%
Forest	98.7%	99.4%	99.1%	62.5%	97.9%	76.3%	96.4%	95.6%	96.0%
Glacier	93.9%	82.8%	88.0%	78.7%	53.5%	63.7%	89.2%	70.2%	78.5%
Mountain	86.3%	92.2%	89.1%	67.1%	69.5%	68.3%	76.4%	85.0%	80.4%
Sea	95.1%	98.6%	96.8%	76.4%	49.4%	60.0%	84.7%	88.2%	86.5%
Street	93.3%	94.4%	93.8%	68.1%	64.7%	66.3%	82.8%	89.4%	86.0%

V. DISCUSSION AND SUMMARY

A. Isolating Pretraining Quality

The comparison between *pretrained-frozen* (PT-F) and *random-frozen* (R-F) directly isolates the contribution of pre-trained feature quality, as both configurations share identical trainable capacity and parameter freezing boundaries. As shown in Table I, PT-F achieves a massive performance lead over R-F, with a test accuracy gap exceeding 50% on average. Because θ_f in R-F is randomly initialized and never updated, it generates low-level representations that are arbitrary and noise-like. The trainable block ‘layer4’ is forced to map these arbitrary patterns into class predictions, severely limiting generalization. In contrast, the ImageNet pre-trained weights in PT-F furnish rich, highly organized, and generalizable features (such as edges, textures, and shape boundaries) that translate seamlessly to the natural scene categories of the Intel dataset.

B. Isolating Trainable Capacity

The comparison between *random-frozen* (R-F) and *random-full* (R-FL) isolates the effect of network capacity, as both start from random weight initializations. R-FL gains a significant advantage by allowing all parameters (including θ_f) to adapt. This permits the early layers to optimize from raw pixel values to fit the target data distribution, creating useful, specialized filters. As a result, R-FL achieves a mean accuracy of 84.69%, compared to the restricted 66.54% of R-F.

However, R-FL takes substantially longer to train (converging in mean 15.0 epochs compared to 8.7 for R-F) and still falls short of the pre-trained PT-F model by a large margin. This indicates that while optimizing trainable capacity helps, it is insufficient to overcome the absence of high-quality initialization when dataset size and training epochs are bounded.

C. Class-wise Confusions

Our test confusion matrices show consistent patterns of category-specific confusion across all conditions, though the errors are far more pronounced in the random initialization models. The primary confusion occurs between:

- **Glacier vs. Mountain:** Visual features such as white snow, jagged ridges, and rock textures are shared heavily between these classes.
- **Buildings vs. Street:** Images of streets frequently contain buildings in the background, and vice-versa, making the spatial context highly overlapping.

These confusions suggest that classification errors are primarily driven by semantic and visual ambiguities inherent to the dataset rather than flaws in the training configuration.

D. Summary

In this study, we conducted a rigorous, capacity-matched evaluation of transfer learning and training from scratch. Our experiments successfully isolated the effects of pre-training and capacity. We showed that pre-trained features provide an enormous boost in final generalization accuracy ($> 50\%$) and converge significantly faster under early stopping. For computer vision applications with restricted datasets or compute constraints, leveraging transfer learning remains the most effective strategy.

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